CUBIQO PHASE 1 MASTER DOC

Pre-RGY Conversation to RGY Capsule

This is the scoped Phase 1 architecture. It covers the landing conversation, anonymous and signed-in memory, consent, perception, user understanding, safety, classification, and creation of an RGY capsule when a real signal emerges. It stops before execution workspaces, publishing, workflow automation, and commercial systems.

Phase 1 only | **No autonomous execution** | **No execution workspace** | **No commercial build** | **Capsule-ready boundary**

1. **Before a goal exists**
   CubiQo is conversational: it listens, responds, remembers safely, and learns the user.

2. **When a signal appears**
   Understanding/classification decides whether this is chat, memory, journal context, or an RGY activity signal.

3. **When strong enough**
   The system writes a structured RGY capsule: color, keyword, domain hint, confidence, status, and user-visible summary.

4. **Boundary**
   Phase 2 will decide what to do with the capsule. Phase 1 only prepares it correctly.

CANONICAL PHASE 1 PIPELINE

Everything before execution

The pipeline is intentionally not "execution first." It is understanding first. A goal or capsule is an outcome of the conversation, not a requirement for every message.

Input

Text, voice, image, camera frame, browser/screen context when user provides it.

Permission

Mic, camera, memory, location, push and tracking consent are checked before use.

Perception

Speech-to-text, vision, OCR-like visible text, screenshot understanding.

Context

Session, time, language, anonymous memory, user profile, journal, signed-in memory.

Understand

Intent, emotional tone, topic, goal hint, user preference, risk, and freshness need.

Safety

Input safety, age-gated/hard-block/crisis checks, output safety for sensitive paths.

Classify

Conversation, memory update, journal insight, onboarding fact, or RGY signal.

Capsule

If useful: save RGY capsule with confidence and status. Phase 1 ends here.

VISUAL FLOW

Three user states in one Phase 1 map

This is the relaxed board version: first-time anonymous user, returning anonymous user, and signed-in engaged customer all flow into the same understanding and capsule boundary.

USER INTERACTION AT LANDING APP

FIRST-TIME ANONYMOUS USER

NO SERVER MEMORY START LOCAL SESSION ONLY

localStorage session_id, first_seen_at, visit_count, language hint. No DB write.

RETURNING ANONYMOUS WITHIN TTL

CUBIQO REMEMBERS LIGHTLY LIKE A FRIEND, NOT A DATABASE

Use local signals and summary only. Nudge to sign up before memory expires.

SIGNED-IN RETURNING ENGAGED USER

DURABLE MEMORY + JOURNAL PROFILE IS UPDATED OVER TIME

Supabase reads profile, memory_events, user_ai_profile, journal insights, signals.

DB / ENV
LOCAL FILES
RUNTIME CONFIG

APIS
AUTH
VOICE
VISION

UNDERSTANDING + CLASSIFICATION
RGY IS ONE POSSIBLE OUTPUT

DECIDE: CHAT, MEMORY, JOURNAL, ONBOARDING, OR CAPSULE

NORMAL
CONVERSATION

MEMORY
UPDATE
PROFILE HINT

RGY SIGNAL
IF STRONG
ENOUGH

CAPSULE
READY + SAVED
PHASE 1 ENDS

OUTPUT: useful response now, not execution

Guardrails: no payments, no final send/submit/push

USER STATES

What happens for each user type

1. Anonymous first-timer

No account, no durable profile, no server memory. CubiQo starts with a clean conversational opener and creates a temporary local session.

Storage	`localStorage.cubiqo_session` only.
Memory	None at first. Light summary can form inside the browser session.
Goal	Not assumed. CubiQo detects, asks, and learns gradually.
Capsule	Can create a local capsule preview, but durable save requires sign-in.

2. Anonymous returning user

The user comes back before local memory expires. CubiQo can reference the last topic lightly and invite account creation when value is clear.

Storage	Local TTL memory, recent signals, optional last summary.
Greeting	Specific but not creepy: "Still thinking about the store idea?"
Conversion	"Want me to keep this longer?" only after enough value exists.
Migration	On sign-up, local memory migrates into Supabase tables.

3. Signed-in engaged user

Durable memory, profile, journal insights, and prior signals become part of the context bundle before CubiQo answers.

Storage	Supabase: profile, signals, memory events, user AI profile, journal.
Greeting	Text-first greeting; voice only after audio unlock.
Understanding	Language, time, location if permitted, user style, recurring blockers.
Capsule	Strong activity signals become durable RGY capsules.

ENGINEERING LAYERS

What sits where

Frontend

- Landing chat surface.
- Anonymous local session manager.
- Voice input capture and audio unlock gate.
- Image/camera upload frame capture after permission.
- Right-side RGY preview panel only when a signal exists.
- Sign-up migration trigger for local memory.

API Routes

- `/api/agent/stream` - conversational answer with context bundle.
- `/api/agent/greet` - text greeting; voice after browser unlock.
- `/api/agent/onboard` - light profile bootstrap for signed-in users.
- `/api/rgy/classify` - safety plus understanding plus RGY signal detection.
- `/api/perception/vision` - structured visual context from image/screenshot.
- `/api/memory/session-close` - writes summaries and commitments.

Supabase Tables

- `profiles` - display name, language, timezone, consent hints, onboarding state.
- `signals` - RGY activity signals and capsule source records.
- `memory_events` - session summaries, commitments, blockers, outcomes, episodes.
- `user_ai_profile` - CubiQo's inferred understanding with confidence and corrections.
- `journal_entries` - user-authored journal plus extracted memory insights.
- `user_permissions` - mic, camera, memory, location, push statuses.
- `safety_patterns` and `system_prompts` - config-driven safety and behavior.

Config and Env

- `OPENAI_API_KEY` - conversation, structured understanding, vision, STT as configured.
- `ELEVENLABS_API_KEY` - optional TTS after user interaction unlock.
- `NEXT_PUBLIC_SUPABASE_URL` and keys - auth and DB access.

- `NEXT_PUBLIC_BASE_URL` - required; never silently fallback to production.
- `RGY_MODEL`, `OPENAI_MODEL`, `UTILITY_MODEL` - runtime model routing.
- `HISTORY_WINDOW_SIZE`, `MEMORY_EVENT_LIMIT` - tunable limits.

MEMORY MODEL

User memory before and after sign-up

This is the center of Phase 1. CubiQo needs a trustworthy understanding of the user before it can safely route goals in later phases.

Memory lane	Anonymous	Signed-in	Update rule
Session memory	localStorage with short TTL	Conversation/session state plus DB summary	Updated during chat; summarized at session boundary
RGY signals	Local preview only	`signals` rows	Written when a strong activity signal appears
Memory events	Not server-side until signup	`memory_events`	Commitments, blockers, outcomes, pivotal episodes, session summaries
User AI profile	Not durable	`user_ai_profile`	Bootstrapped on onboarding; refreshed after meaningful sessions
Journal	Not durable	`journal_entries`	User-authored; extracted into memory only with consent and care

UNDERSTANDING AND CLASSIFICATION

RGY is an output, not the whole brain

Conversation

Normal chat stays normal chat. CubiQo answers, asks clarifying questions, reflects preferences, and may update light memory.

No capsule required

Memory update

If the user reveals stable facts, preferences, commitments, or blockers, CubiQo records them with confidence and source.

Profile + memory

RGY capsule

If a meaningful activity signal emerges, CubiQo writes color, keyword, intent hint, safety state, confidence, and summary.

Capsule-ready

Classification contract

Safety always runs first. Then CubiQo decides whether the input is ordinary conversation, onboarding material, a memory event, a journal insight, a question needing fresh context, or a true RGY signal. Only the last path creates a capsule.

GUARDRAILS

The two hard rules still apply in Phase 1

1. Do not touch payments

Phase 1 does not initiate purchases, billing, checkout, payment setup, or financial transactions. It can discuss concepts but does not act.

2. Do not final-submit anything

No autonomous send, submit, publish, push, deploy, or final external action. The user keeps the last button in later phases too.

Safety layers included here

Input safety, output safety for sensitive outputs, consent checks before sensors, RLS for user-owned rows, and no silent global connector fallback. These are Phase 1 foundations because trust starts before any external execution exists.

PHASE 1 BUILD SCOPE

What to build now

Must build

- Anonymous local memory with TTL and sign-up migration.
- Signed-in onboarding conversation.
- `memory_events` and `user_ai_profile` tables with RLS.
- `user_permissions` table for mic, camera, memory, location, push.
- Text-first greeting and audio unlock before TTS playback.
- Reliable STT path and optional Web Speech fallback.
- Vision route returning structured visual context, not just a caption.
- Understanding/classification route with RGY as one possible result.
- Capsule save boundary: color, keyword, confidence, status, summary.

Explicitly not Phase 1

- Task execution and route planning.
- Structured execution workspace.
- Domain-specific execution flows.
- Browser automation for acting on external sites.
- Publishing, submitting, deploying, or final-button actions.
- Commercial systems, checkout, and revenue analytics.
- Market watch, background briefings, intervention systems.

MASTER CHECKLIST

Phase 1 done means this works

Area	Definition of done
Anonymous first visit	New local session created. No DB write. Neutral greeting. User can chat normally.
Anonymous return	Within TTL, CubiQo references prior topic lightly. After TTL, clean slate.
Sign-up migration	Local signals and summary migrate into Supabase. localStorage clears after successful write.
Onboarding	Signed-in user gets short conversational onboarding. Primary goal and initial user_ai_profile are written.
Memory	Session summaries, commitments, blockers, journal insights, and pivotal episodes are saved with source and confidence.
Permissions	Mic/camera/location/memory preferences are recorded and respected before each use.
Understanding	System separates ordinary chat, memory updates, journal insights, onboarding facts, and RGY signals.
RGY capsule	Strong signal creates durable capsule record with color, keyword, intent hint, confidence, and summary.
Safety	Input safety, output safety, RLS, no silent fallback, no payments, and no final submissions verified.

COVERAGE AUDIT

Does this cover the full Phase A plan?

Yes as a Phase A master, with one important boundary: the main body explains the foundation from landing conversation to capsule readiness. This checklist captures the rest of Phase A as implementation scope, while explicitly parking subscription, Tier 4 affiliate networks, browser-extension attribution, and Amazon PA API for later.

Area from the larger plan	Status in this master doc	Why
Input: voice, text, image, camera, browser context	Covered	Included as the Phase 1 perception layer, with permissions before sensors.
Perception layer: STT, vision, screenshot understanding	Covered	Included, with structured visual context as a target rather than plain captions.
Context layer: time, language, location, memory, user profile	Covered	Included as the context bundle for signed-in and anonymous states.
Daily context / fresh world state	Phase A expansion	Profile slots and context concept are included; cron-backed digest/source tracking becomes an expansion item.
RGY classification	Covered	Reframed correctly: RGY is one output of Understanding + Classification, not every message.
Signals and capsule creation	Covered	Phase 1 ends at a saved, structured capsule with color, keyword, status, summary, and confidence.
Memory events and AI-maintained user profile	Covered	Included as core Phase 1 tables and behavior.

Area from the larger plan	Status in this master doc	Why
Journal influence on user understanding	Covered	Included as a memory lane, with extracted insights feeding memory only with care.
Anonymous local memory and sign-up migration	Covered	Included for first-time, returning, and signed-in user flows.
Onboarding conversation	Covered	Included as a required signed-in bootstrap path, not a form.
Permissions and consent	Covered	Included as a first-class layer and Supabase table.
Input and output safety	Covered	Included as Phase 1 trust foundations.
Audio greeting and ElevenLabs	Partial	Text-first greeting and audio unlock are included. Voice generation implementation belongs in build tickets.
Rate limiting and API usage cost protection	Phase A expansion	Cost protection is part of Phase A. Paid subscription packaging remains later.
Background briefings, interventions, market watch, cron refreshes	Phase A expansion	These sit after capsule creation, so they are not in the pre-capsule visual but are included in the Phase A checklist.
Execution workspace / dashboard after capsule	Later	The workspace that acts on capsules should be a separate Phase B document.
Commerce recommendations, cards, coupons, tracked links	Phase A expansion	Include quality gate, transparent cards, Tier 1 partners, coupons/rates, and click tracking after memory is trustworthy.
Direct partners, broad networks, product APIs, postbacks	Mixed	Tier 1 direct partners and postbacks can be Phase A. Tier 4 networks and Amazon PA API are later.
Subscription checkout and paid tiers	Later	Explicitly parked for later, per current strategy.
Admin monetization dashboard	Phase A expansion	Basic admin/cost/recommendation metrics are useful once cards and tracking exist.

Area from the larger plan	Status in this master doc	Why
GDPR export/delete, offline fallback, PWA/mobile hardening	Partial	The trust need is noted. Full product/legal implementation should become separate tickets.

PHASE A CHECKLIST

Master implementation checklist

This is the end-to-end checklist for Phase A. Items marked "Later" are deliberately not part of Phase A execution: subscription checkout, Tier 4 affiliate networks, browser-extension attribution, and Amazon PA API.

Workstream	Phase A scope	Status
Anonymous first visit	Create `cubiqo_session` locally, no DB write, neutral greeting, clean state.	Core
Anonymous return	Use local TTL memory, update visit count, greet with last topic when appropriate, expire after TTL.	Core
Anonymous conversion	Trigger "want me to keep this?" only after enough value: repeated exchanges, green signal, or commitment.	Core
Sign-up migration	Migrate local signals and summaries into `signals`, `memory_events`, and `profiles`.	Core
Onboarding	Three-question conversational bootstrap that writes primary goal, goal domain, initial memory, and user AI profile.	Core
Perception	Text pass-through, reliable STT, image/camera/screenshot to structured visual context.	Core
Permissions	Track mic, camera, memory, location, push, and tracking consent in `user_permissions`.	Core
Context assembly	Build context from time, timezone, language, location if allowed, memory, journal, profile, and recent signals.	Core
Understanding + classification	Decide between chat, memory update, journal insight, onboarding fact, fresh context question, or RGY signal.	Core
RGY capsule	Create durable capsule when signal is strong: color, keyword, intent hint, confidence, safety state, summary.	Core
Memory architecture	Create `memory_events`, add keywords, confidence, source, user confirmation, archived_at, and episode	Core

Workstream	Phase A scope	Status
	support.	
User AI profile	Create `user_ai_profile` with inferred fields, confidence map, corrections, open loops, and session count.	Core
Journal memory bridge	Extract commitments, blockers, goals, and mood signals from journal entries into memory with user-respecting rules.	Core
Multilingual	Store `language_preference` and `script_preference`; match Hinglish, Spanglish, and language switching.	Core
Voice output	Text first; voice after user interaction unlock; no silent autoplay dependency.	Core
Safety	Input safety, output safety, crisis/hard-block/age-gate patterns, RLS, and no unsafe global fallbacks.	Core
Guardrails	No payments. No autonomous final send, submit, publish, push, deploy, or external final action.	Core
Usage and cost protection	Create `user_api_usage` and `api_usage_events`; enforce fair-use/rate limits without subscription checkout.	Expansion
Daily context	Add generated_at/expires_at/source-backed daily context. Use live search only when user asks current/latest/news/price/law.	Expansion
Background capsules	After green signal, create/update `capsule_briefings`, retry failures, and source research safely.	Expansion
Interventions	Use commitments, stale goals, inactivity, open loops, and market watch to suggest timely nudges without over-pushing.	Expansion
Job/cron reliability	Add `job_runs` locks/idempotency, retry queues, cron health, and duplicate-prevention.	Expansion
Recommendation quality gate	Only show a card when intent and relevance are high, no better free option is hidden, and disclosure is clear.	Expansion
Transparent recommendation cards	Support Get started, Save, I already have this, Find cheaper, Compare options, Tell me more, and visible disclosure.	Expansion

Workstream	Phase A scope	Status
Tier 1 direct partners	Seed curated partners such as Shopify, Printify, Fiverr, HubSpot, Canva, Notion, Coursera, Udemy, Webflow, and similar.	Expansion
Coupon and rate search	Compare options, fetch better rates, find working coupons, and explain why a recommendation is useful.	Expansion
Tracking and attribution	Use `recommendation_events`, `affiliate_clicks`, `saved_recommendations`, promo codes, postbacks, and self-reporting.	Expansion
Admin/ops visibility	Track DAU, signal count, memory writes, card CTR, saves, bad recommendations, API cost, cron health, and route errors.	Expansion
GDPR/export/delete	Provide user export, account delete, local session clearing, and minimal sensitive logging.	Expansion
Error and offline states	Handle Supabase/OpenAI/network failures gracefully; queue unsent local messages where safe.	Expansion
PWA/mobile hardening	Add install prompt, mobile audio gesture handling, camera facing-mode, and push constraints.	Expansion
Subscription checkout	Stripe checkout, paid tiers, and subscription packaging.	Later
Tier 4 affiliate networks	Impact/CJ/ShareASale-style broad network expansion.	Later
Browser-extension attribution	Extension-based checkout/domain attribution.	Later
Amazon PA API	Amazon product cards, ASIN lookup, price/review metadata.	Later

SINGLE SOURCE

How to use this doc

This file is now the Phase A master visual and engineering scope: `docs/cubiqo-capability-engine-future-state.html`. Use the top sections for the clean pre-capsule architecture and the bottom checklist for the full Phase A implementation map.